contract has taken over. This is the default way of looking at continuous futures time series in many market data applications and if you, for instance, chart the c1 codes in Reuters, this is what you get. In this example, it is easy to see right away when the contract rollovers occurred, even without the circles that I put in. The seemingly erratic behaviour of the price during these periods does not at all reflect the actual market conditions at the time and basing your simulations on such data will produce nonsense results.

Now compare this with the more normal-looking price curve in Figure 2.6. Notice how it is no longer possible to see where the contract rollovers occurred and the artificial gaps have been removed. If you look even closer, you will notice that while the final price is the same in the end, there are significant price differences between the two series on the left-hand side of the x-axis. Whereas the unadjusted peak reading in October was about 17.3, the adjusted chart shows a peak of over 30. The difference can occur in either direction, depending on whether there is a positive or negative basis gap at the time of the roll.

The reason for this price discrepancy in the past prices is that I use back-adjusted price charts here. For a back-adjusted chart, the current price is always correct at the right-hand side of the series, but all previous contracts will have a mismatch. When the roll occurs, the back-adjusted series will adjust all series back in time and remove the artificial gap. This means that the whole time series back in time will have to be shifted up or down to match the new series.

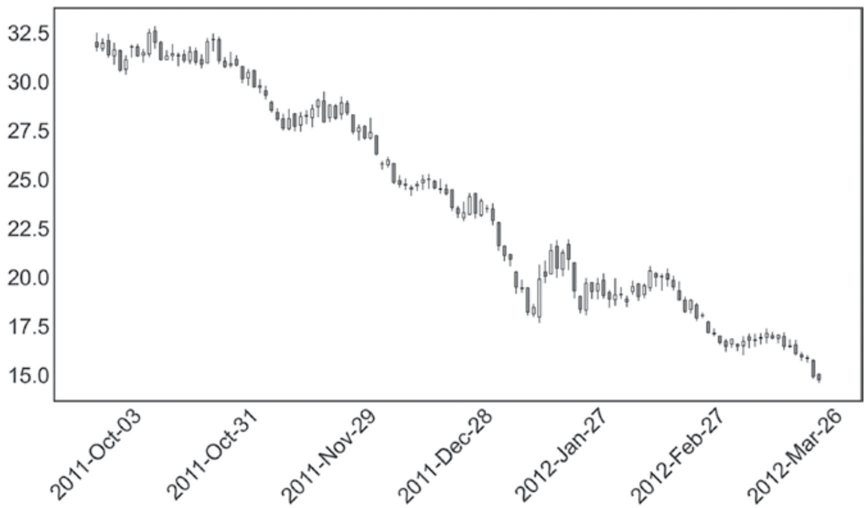

FIGURE 2.6 Properly adjusted time series for rough rice.